Week 9
Multiple linear regression

Applied Statistical Analysis/Quantitative Methods I

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Why do we need multiple linear regression?

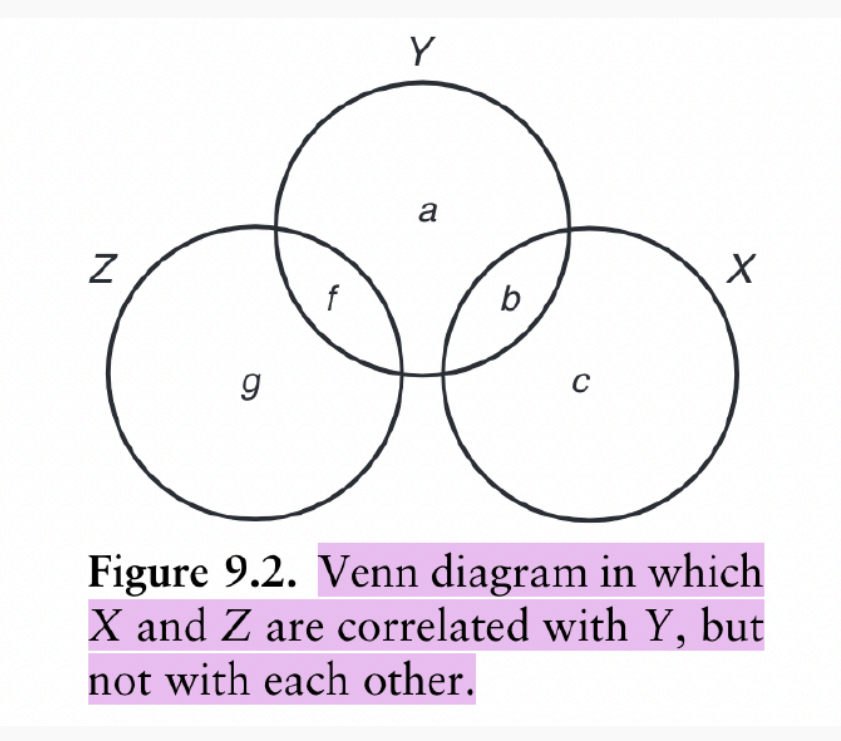


Figure 9.2. Venn diagram in which X and Z are correlated with Y, but not with each other.

"In that case — which, we have noted, is unlikely in applied research — we can safely omit consideration of Z when considering the effects of X on Y. In that figure, the relationship between X and Y (the area b) is unaffected by the presence (or absence) of Z in the model."

Kellstedt and Whitten (2018, 213)

Why do we need multiple linear regression?

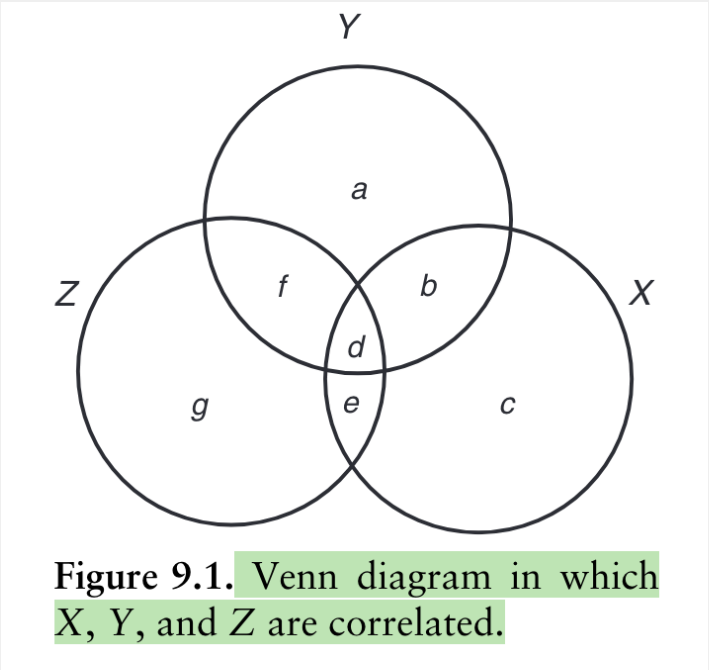


Figure 9.1. Venn diagram in which X, Y, and Z are correlated.

"If, hypothetically, we erased the circle for Z from the figure, we would (incorrectly) attribute all of the area b + d to X, when in fact the d portion of the variation in Y is shared by both X and Z. This is why, when Z is related to both X and Y, if we fail to control for Z, we will end up with biased estimates of X's effect on Y"

Kellstedt and Whitten (2018, 212)

Ex: Multiple linear regression

Does time spent working take away from political knowledge?




```
Call:
lm(formula = polknow ~ work_hours, data = samp)

Residuals:
Min      1Q Median      3Q     Max
-7.686 -1.760 -0.061  1.683 10.385

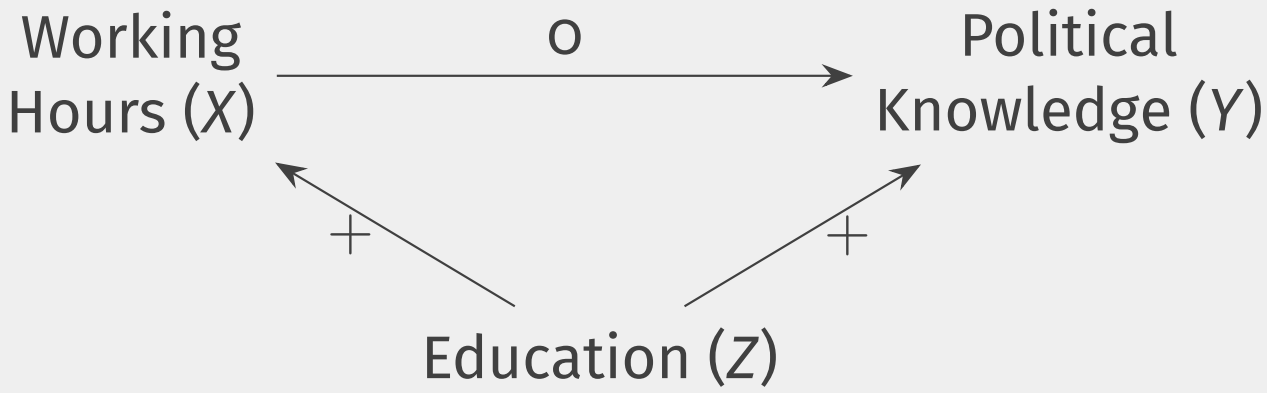
Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)  14.59166    1.09142  13.369   <2e-16 ***
work_hours    0.06791    0.02640   2.572    0.0103 *
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.565 on 998 degrees of freedom
Multiple R-squared: 0.006585, Adjusted R-squared: 0.00559
F-statistic: 6.615 on 1 and 998 DF, p-value: 0.01025
```

How convincing is this finding?

Ex: Multiple linear regression

Does time spent working take away from political knowledge?



- Education as confounder: Controlling for education is relevant because it might drive both working hours and political knowledge
  - Education impacts (i.e., "causally prior to") working hours

$\rightarrow$ Avoid omitted variable bias by including relevant control variables (Z) that are correlated with both X and Y

Adding Covariates

Multiple regression analysis allows us to add covariates $X_2, \ldots, X_k$ on top of X_1 in a regression of Y:

$$Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k + \epsilon$$

- Need additional covariates to minimize prediction error $(\widehat{Y} - Y)$
  - Need to adjust for additional covariates to minimize bias $(\widehat{\beta_1} - \beta_1)$

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- Estimation is same for both purposes; rationale underlying model specification that changes

Multiple Linear Regression Model

- General multiple linear regression model:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k + \epsilon$$

  - ϵ contains factors other than $X_1, \ldots, X_k$ that affect Y
  - We assume all factors in unobserved error term are uncorrelated with explanatory variables

- Estimation approach is same as in two-variable case, i.e., to minimize sum of squared residuals (but now we get $k + 1$ normal equations):

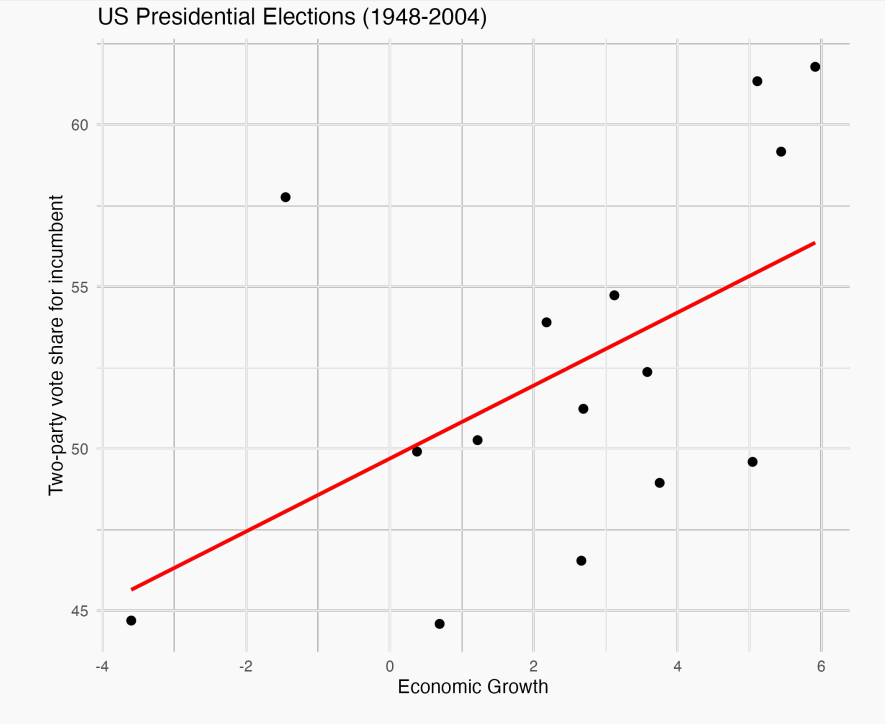
$$\underset{\widehat{\alpha},\widehat{\beta}_1,\ldots,\widehat{\beta}_k}{\text{minimize}} \quad \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} \left(y_i - \widehat{\beta_0} - \widehat{\beta}_1 X_{i1} - \cdots - \widehat{\beta}_k X_{ik} \right)^2$$

Interpreting the Coefficients

- Interpretation of regression coefficients:
  - $\widehat{\beta_0}$ Predicted value of Y when all X's equal zero
  - $\widehat{\beta_1}$ On average, a one-unit change in X_1 leads to a $\widehat{\beta_1}$-unit change in Y, holding everything else constant
  - $\vdots$
  - $\widehat{\beta_k}$ On average, a one-unit change in X_k leads to a $\widehat{\beta_k}$-unit change in Y, holding everything else constant
- Remember: In $k + 1$-dimensional space, a fitted multiple regression model no longer defines a line, but a *hyperplane*
  - So, for $k = 2$, OLS means fitting a least squares plane that best fits cloud of data points in a three-dimensional space

An example

- Let's look at US presidential election example again
- We used economic growth to predict two-party vote share:



- Can you think of an alternative explanation for success of presidential party?

EX: US Presidential Elections

- Let's consider presidential popularity + economic growth
- Popularity for presidents prior to election ranged between 31% and 74%
- Our model becomes:

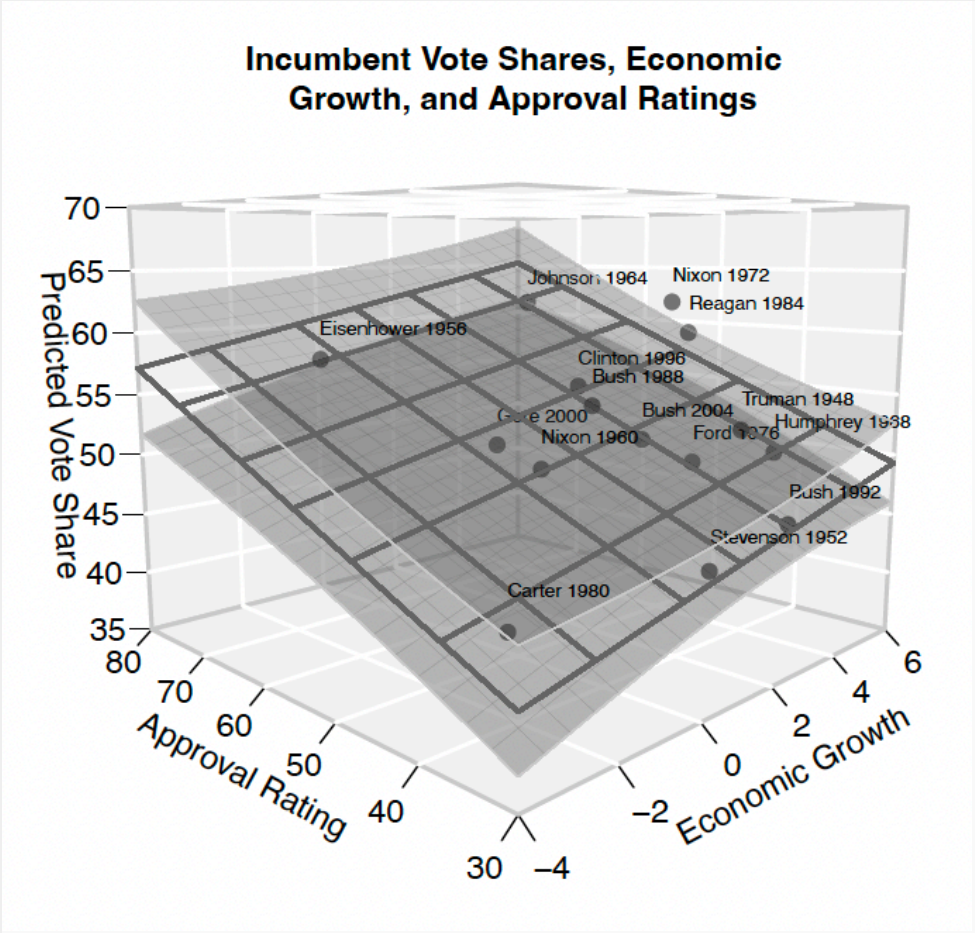
$$\text{VoteShare} = \alpha + \beta_1 \text{Growth} + \beta_2 \text{Approval} + \epsilon$$

- Results of regression of vote share on growth and approval rating:

Variable	Estimate	SE
Constant	34.83	2.77
Growth	0.81	0.27
Approval	0.32	0.06
$R^2 = 0.81$		
Obs. = 15		

Visualization of example

$$\widehat{\text{VoteShare}} = 34.83 + 0.81 \times \text{Growth} + 0.32 \times \text{Approval}$$



Statistical Control

So what is *statistical control* and how do we get an effect of ‘Approval’ of 0.32 independent of ‘Growth’?

Statistical Control

- We control for ‘Growth’ by removing its effect from ‘VoteShare’ and ‘Approval’ before regressing them

Statistical Control

- We control for 'Growth' by removing its effect from 'VoteShare' and 'Approval' before regressing them
  - What is effect of approval rating on vote share given growth?

1. Let's start by running the simple regression
   $\text{VoteShare} \sim \text{Growth}$

DV: Vote Share	Estimate	SE
Constant	49.70	1.75
Growth	1.13	0.49



- Residuals in this model are part of 'Vote Share' unexplained by 'Growth'
- In other words, we remove effect of growth from vote share variable

Statistical Control

- Now, we remove the effect of 'Growth' (X_1) from 'Approval' (X_2)

2. Let's regress $\text{Approval} \sim \text{Growth}$

DV: Approval	Estimate	SE
Constant	46.54	4.67
Growth	0.98	1.32



- Residuals in this model are part of 'Approval' unexplained by 'Growth'
- In other words, we remove effect of growth from approval rating

Statistical Control

- Now, we regress residuals from step (1) on residuals from step (2):

DV: Residuals 1	Estimate	SE
Constant	0.00	0.66
Residuals 2	0.32	0.06



- We "controlled" for effect of Growth (removed from both vote share and approval), can compute effect of approval on vote share

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- Gives exactly same coefficient (and SE) as a multiple regression:

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- Multiple regression coefficients can statistically control for (be independent of) effects of other variables

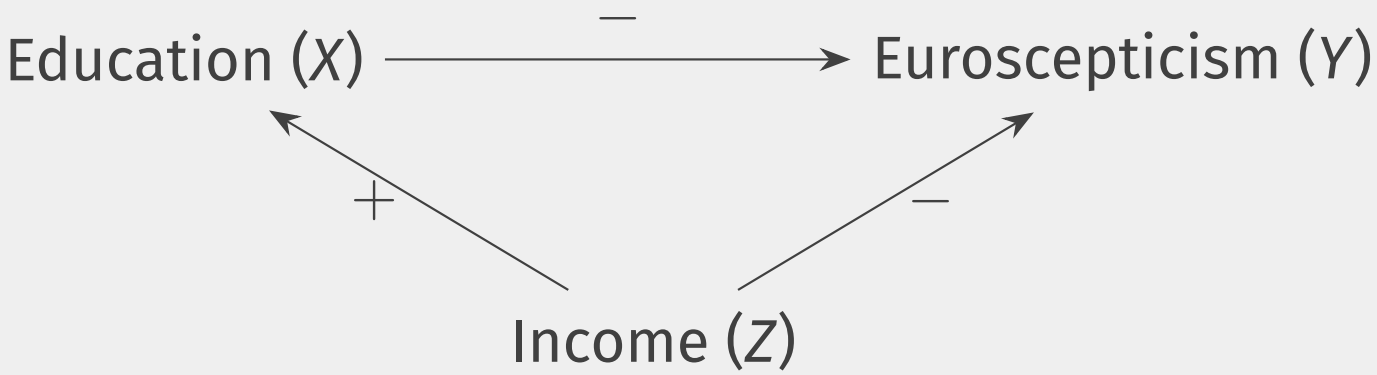
Tutorial exercises

What is relationship between education and Euroscepticism?



Hypothesis 1: Greater years of education $\rightarrow$ lower level of Euroscepticism

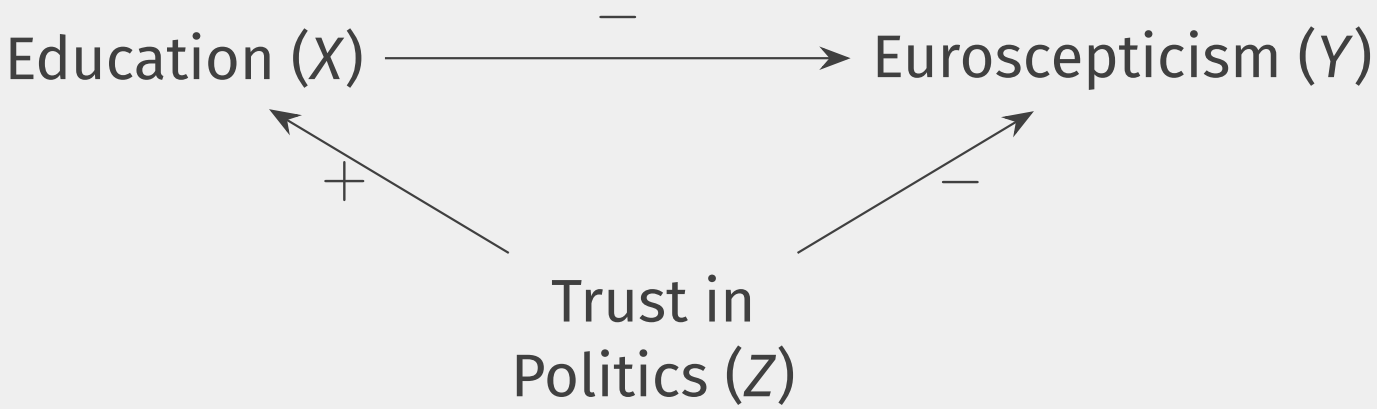
Relationship of education & Euroscepticism?



Hypothesis 2: Higher income $\rightarrow$ lower level of Euroscepticism

$\rightarrow$ Economic dimension

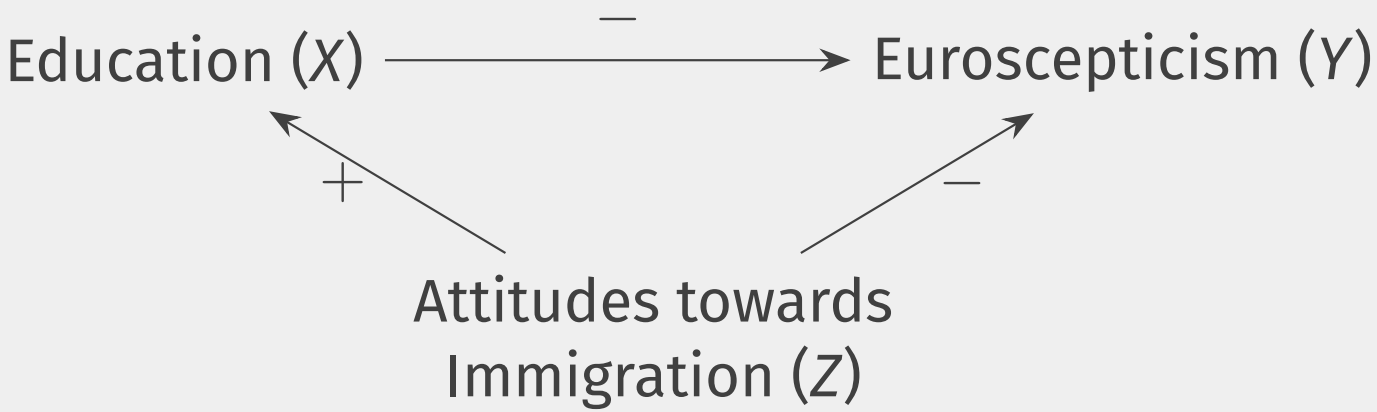
Relationship of education & Euroscepticism?



Hypothesis 3: Higher trust in politics $\rightarrow$ lower level of Euroscepticism

$\rightarrow$ Political dimension

Relationship of education & Euroscepticism?



Hypothesis 4: More positive attitudes towards immigration → lower level of Euroscepticism

→ Cultural dimension

Class business

- Read required (and suggested) online materials
- Finish Problem set #3!